

**ORIGINAL ARTICLE****Effect of Different Smoking Methods on the Fatty Acid Composition of Oils
Extracted from Two Fish Species (*Cyprinus carpio* and *Clarias gariepinus*) Captured
from Fish Ponds in Batié, West-Cameroon**

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Funding source

None.

Abstract

The extension of the shelf-life of food in order to ensure availability also requires techniques that preserve nutrients. Amongst the preservation methods used in fish preservation, smoking is the most famous one and is still widely used in Africa. Different smoking methods are applied and influence fish quality differently. The work was conducted in order to evaluate the effect of different smoking methods applied in Cameroon on the fatty acid composition of oil extracted from two fish (*Cyprinus carpio* and *Clarias gariepinus*). The fish were captured from ponds and taken to the laboratory in Ice. Fresh fish served as control. The other fish were separated into parts, eviscerated and non-eviscerated. They were smoked using improved and traditional smoking houses with three types of wood (*Mangifera indica*, *Rhizophora mangle*, *Psidium guajava*), some with filter and others without filter. Oils were extracted from fish samples using the Bligh and Dyer method and the obtained oils analyzed for their fatty acid compositions. Results showed that all the smoking methods significantly ($p < 0.05$) increase the saturated and monounsaturated fatty acid content of these fish while decreasing the polyunsaturated ones. Treatments T2, T7, T3' and T11' had the best linoleic acid contents. Treatments T1' T2', T5', T6' T8', T9', T10', T12', T13' and T14' exhibited the highest docosahexaenoic acid contents in the oil from *Cyprinus carpio* while similar observation was made with treatments T9, T10, T11, T13 and T14 in *Clarias gariepinus* oil. The above mentioned treatments should be applied for the preservation of these fish and the maintenance of the quality of their unsaturated fatty acids.

Keywords: Smoking; fatty acid composition; *Cyprinus carpio* *Clarias gariepinus*

Abbreviations

T0' = Control; T1' = Non-eviscerated *Cyprinus carpio* fish smoked with guava tree without filter ; T2' = Non-eviscerated *Cyprinus carpio* fish smoked with guava tree with filter ; T3' = Eviscerated *Cyprinus carpio* fish smoked with guava tree without filter ; T4' = Eviscerated *Cyprinus carpio* fish smoked with guava tree with filter ; T5' = Eviscerated *Cyprinus carpio* fish smoked with mangrove tree without filter ; T6' = Eviscerated *Cyprinus carpio* fish smoked with mangrove tree with filter ; T7' Non-eviscerated *Cyprinus carpio* fish smoked with mangrove tree without filter ; T8' = Non-eviscerated *Cyprinus carpio* fish smoked with mangrove tree with filter ; T9' = Non-eviscerated *Cyprinus carpio* fish smoked with mango tree without filter ; T10' = Non-eviscerated *Cyprinus carpio* fish smoked with mango tree with filter ; T11' = Eviscerated *Cyprinus carpio* fish smoked with mango tree without filter ; T12' = Eviscerated *Cyprinus carpio* fish smoked with mango tree with filter ; T13' = Non-eviscerated *Cyprinus carpio* fish smoked using the traditional method ; T14' = Eviscerated *Cyprinus carpio* fish smoked using the traditional method



T0 = Control ; T1 = Non-eviscerated *Clarias gariepius* fish smoked with guava tree without filter; T2 = Non-eviscerated *Clarias gariepius* fish smoked with guava tree with filter; T3 = Eviscerated *Clarias gariepius* fish smoked with guava tree without filter; T4 = Eviscerated *Clarias gariepius* fish smoked with guava tree with filter ; T5 = Non-eviscerated *Clarias gariepius* fish smoked with Mangrove tree without filter ; T6 = Non-eviscerated *Clarias gariepius* fish smoked with Mangrove tree with filter ; T7 = Eviscerated *Clarias gariepius* fish smoked with Mangrove tree without filter ; T8 = Eviscerated *Clarias gariepius* fish smoked with Mangrove tree with filter ; T9 = Non-eviscerated *Clarias gariepius* fish smoked with Mango tree without filter ; T10 = Non-eviscerated *Clarias gariepius* fish smoked with mango tree with filter ; T11 = Eviscerated *Clarias gariepius* fish smoked with mango tree with filter ; T12 = Eviscerated *Clarias gariepius* fish smoked with Mango tree with filter ; T13 = Non-eviscerated *Clarias gariepius* fish smoked using the traditional method ; T14 = Eviscerated *Clarias gariepius* fish smoked using the traditional method

1. Introduction

Fish is a very important food though it constitutes an extremely perishable commodity (Ajonina, 2022). In 2016, the global fish production reached a record value of about 171 million tons. This resulted to an increase in the per capita consumption of fish from 11.3 kg/person in 1980 to 20.5 kg/person in 2017 (FAO, 2018). Fish constitutes a major source of animal protein, accessible to low-income households (Ekomy *et al.*, 2013). However, the lack of facilities for its preservation leads to high losses which make the fish to be unavailable in quantity to ensure food security, hence the need to master conservation techniques (Daniel, 2013). The most used preservation techniques for fish are drying and smoking. During smoking, the fish absorbs volatile compounds from the smoke which give them particular colors and pleasant flavors (Ajonina, 2022). Tiwo *et al.* (2019) evaluated the effect of different smoking processes on the proximate composition of *Clarias gariepinus* and *Cyprinus carpio*. They revealed that these fish are very rich in proteins, about $86.56 \pm 3.09\%$ and $88.65 \pm 1.54\%$ respectively for *Clarias gariepinus* and *Cyprinus carpio*. The lipid contents obtained were found to be $10.46 \pm 0.13\%$ and $5.46 \pm 0.11\%$ respectively for *Clarias gariepinus* and *Cyprinus carpio*. They recorded a significant increase ($p < 0.05$) in the lipid content of the fish after smoking. Fish smoked with *Psidium guajava* without filter exhibited higher lipid contents, 14.17 ± 0.15 and 14.96 ± 0.05 respectively for non-eviscerated *Clarias gariepius* fish smoked

with guava tree without filter (T1) and for Eviscerated *Clarias gariepius* fish smoked with guava tree without filter (T3).

Traditional production units do not also always lead to good quality products. They were found to contain significant level of polycyclic aromatic hydrocarbons, especially benzo (a) pyrene which has harmful effects on the health of those involved in smoking and consumers (Nsoga, 2021). As a result, the problem related to the quality of smoked fish remains relevant. It is therefore important to find new effective methods to improve the nutritional quality of smoked fish. To this end, scientists have made appreciable progress in improving smoked products. Ljubojevic *et al.* (2016) showed that smoking increases the lipid content of carp by 28%. In the same line, Kafeelah *et al.* (2015) reported that the exposure duration to smoke causes certain changes in the sensory properties of finished products and increases the levels of polycyclic aromatic hydrocarbons in fish muscles. However, very few data are available of the effect of traditional smoking methods on the fatty acid composition of fish. Taking into account these observations, smoking with different types of woods and smoke houses can differently affect the quality of the fish, especially its fatty acid composition. This study was conducted in order to evaluate the effect of different smoking methods on the fatty acid composition of oils extracted from two fish Species (*Cyprinus carpio* and *Clarias gariepinus*) Captured from Fish Ponds in Batié.

2. Materials and methods

2.1. Animal material

Cyprinus carpio and *Clarias gariepinus* were collected in ponds from a fish farmer in the Batié locality, West-Region of Cameroon. These fish weighing between 400 and 600 g were preserved in ice (1:1 ratio to maintain freshness) and transported directly to the smoking site in Yabassi, Nkam Division, Littoral Region of Cameroon.

2.2. Plant material

Mangifera indica, *Psidium guajava* and *Rhizophora mangle* were used for smoking. These woods were previously dried under the sun for few days before the start of the smoking process. They were dried until stable masses were obtained, this in order to eliminate the effects of humidity during smoking.

2.3. Smoking process

Smoking consisted of exposing the whole fresh fish, gutted or not, to the action of the smoke produced by the combustion of previously selected woods. Some of these treatments were carried-out with a filter. The smoking process was carried out for 7 hours. The improved smokehouses were designed to prevent fish from burning or falling. The filters were mounted under the last rack 25 cm from the one in contact with the fish. The metallic filter used consisted of a 100 µm mesh screen mounted on a rack.

2.4. Determination of fatty acid composition

2.4.1. Extraction of oils from different samples of smoked fish

The oils were extracted from the different samples using the [Bligh & Dyer \(1959\)](#) method. 100 g of sample was introduced into a glass mixer and 200 ml of the mixture chloroform/methanol (2/1v/v) was added, followed with 80 ml of water.

After grinding for 10 to 15 min, 50 ml of chloroform was added to the mixture and the latter was mixed for approximately 1 min. After filtration, the liquid part was introduced into a separating funnel for phase separation.



Figure 1 (A-C): *Mangifera indica* (Mango)(A); *Psidium guajava* (guava)(B); *Rhizophora mangle* (mangrove) (C)

At the end of the decantation, the lower phase consisting of the chloroform/oil mixture was introduced into a round bottom flask and the solvent evaporated under vacuum at 60°C on a rotatory evaporator R-124 (Büchi Water Bath B 480) to separate the oil from the chloroform. After extraction of the oil from each sample, they were kept in dark bottles (in order to avoid oxidation) and the cakes obtained after extraction of the oil dried in an oven at 45°C for 48 hours

and stored in opaque boxes for the study of the functional properties (proteins).

2.4.2. Determination The fatty acid profile of oils

The fatty acid composition was determined by gas chromatography as described by AOCS (2017).

2.4.2.1. Preparation of fatty acid methyl esters

About 5 ml of NaOH 0.5 N methanolic acid (1 g of NaOH in 50 ml of methanol) was added to 100 mg of fish oil. The mixture was boiled for 5 min and 6 ml of BF₃-methanol (Boron trifluoride-methanol) added into the flask. The mixture was heated under reflux for 5 minutes at 98°C. The flask was cooled and 6 ml of saturated NaCl was added and transferred to the separatory funnel. Then the mixture was filtered using Na₂SO₄ as drying agent. The obtained filtrate was concentrated to 1 ml and used for gas chromatography.

2.4.2.2. Separation by gas chromatography (GC)

The separation of fatty acid methyl esters was carried out by gas chromatography. The GC was programmed for the injection to take place at 265°C. The capillary column (Elite -225; 30 mm, 0.25 mm id, 0.25 µm) was under nitrogen at 0.6 ml/min at 110 °C for 4 min. The temperature was programmed to increase at 2.7°C/min to 240°C and then held for 5 minutes. 1 µl of sample was injected and identified by their retention time using Chromcard software. Each component was identified and quantified from the computer-generated chromatograms against standards analyzed following the same method.

2.5. Statistical analysis

Results were expressed as means ± Standard deviation using the Excel from Microsoft Office version 2020. The obtained data were subjected to one way analysis of variance (ANOVA) with the Bonferroni tests using the software Graphpad-

InStat version 3.05, to evaluate the statistical implication of the data. The variances were important at probability level less than 5%.

3. Results

3.1. Effect of smoking on the fatty acid composition of *Cyprinus carpio*

Table 1 presents the effects of smoking on the fatty acid composition of *Cyprinus carpio* fish. The amount of saturated, monounsaturated and polyunsaturated fatty acids were ranged between 34.68 – 46.11, 26.99 – 34.09 and 20.68 – 35.94% respectively. Generally, the amount of unsaturated fatty acids was higher in this oil compared to the saturated ones. As far as each group of the fatty acid is concerned, palmitic acid was the most represented saturated fatty acid (26.54 – 38.28%), oleic acid the most represented monounsaturated fatty acid (24.41 – 36.24%) and linoleic acid the most represented polyunsaturated fatty acid (14.11 – 34.44%). The effect of smoking on the fatty acid composition of *Cyprinus carpio* oil showed that all treatments (except T3) significantly (p<0.05) increase the content in saturated fatty acid compared to the control. Similar observation was made with monounsaturated fatty acids where all the treatments (except T11') increased in a significant (p<0.05) manner the content in monounsaturated fatty acid compared to the control. The amount of polyunsaturated fatty acids was also found to decrease with treatment in all samples compared to the control. This decrease was more pronounced with T1', T5', T6', T8', T12', T13' and T14' respectively. T7' was the only treated sample with an appreciable amount of eicosapentanoic acid. The concentration of docosahexaenoic acid in this fish was ranged between (0.61-7.71%) which was higher than that of eicosapentaenoic acid (0.00-1.33%). Treatments T1' T2', T5', T6' T8', T9', T10',

Table 1: Effect of smoking processes on the fatty acid composition of *Cyprinus carpio*

	T0'	T1'	T2'	T3'	T4'	T5'	T6'	T7'
C12:0	0.74± 0.04 ^a	1.31± 0.05 ^c	1.34±0.04 ^c	0.89± 0.14 ^a	1.11±0.04 ^b	1.35± 0.03 ^c	1.20± 0.05 ^b	0.89± 0.07 ^a
C13:0	0.04± 0.01 ^a	0.07±0.01 ^b	0.06± 0.01 ^b	0.04± 0.00 ^a	0.06±0.02 ^b	n/a	0.08± 0.02 ^b	0.05± 0.01 ^a
C14:0	2.37± 0.09 ^a	3.41± 0.07 ^c	2.98± 0.09 ^b	2.43± 0.11 ^a	3.01±0.08 ^b	3.81± 0.04 ^c	3.49± 0.02 ^c	2.71± 0.41 ^a
C15:0	0.52± 0.02 ^a	0.95± 0.02 ^b	0.42±0.01 ^b	0.54± 0.05 ^a	0.56± 0.04 ^a	0.76± 0.02 ^d	0.57± 0.01 ^a	0.69±0.08 ^c
C16:0	32.24± 0.81 ^a	37.10±0.41 ^b	31.24± 0.4 ^a	26.54±0.44 ^b	31.41± 0.07 ^a	33.74± 0.41 ^a	36.28±0.05 ^b	31.92± 0.41 ^a
C17:0	0.97± 0.04 ^a	1.89± 0.02 ^d	1.50± 0.04 ^c	1.30±0.04 ^c	0.50± 0.04 ^b	2.78±0.04 ^f	1.84±0.02 ^d	1.63± 0.12 ^c
C18:0	0.72± 0.05 ^a	0.78± 0.04 ^a	0.72± 0.02 ^a	0.89± 0.02 ^c	0.40±0.01 ^b	0.43± 0.02 ^b	0.61± 0.04 ^a	0.74± 0.02 ^a
C19:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
C20:0	n/a	n/a	n/a	0.05± 0.02 ^a	n/a	n/a	n/a	n/a
C21:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
ΣSFA	36.63± 0.01^a	45.51±0.09^d	37.66±0.08^b	34.68± 0.04^c	37.05± 0.13^b	42.87±0.06^e	44.07±0.22^b	38.20±0.05^b
C16:1(n-7)	n/a	0.62±0.01 ^b	n/a	0.16± 0.02 ^a	n/a	n/a	n/a	n/a
C18:1(n-9)	26.59± 1.02 ^a	32.01± 0.05 ^b	31.24±0.12 ^b	28.94± 0.25 ^c	32.36± 0.88 ^b	31.80± 0.04 ^b	32.40±0.01 ^b	31.97± 1.02 ^b
C20:1(n-9)	0.84± 0.09 ^a	1.18± 0.03 ^c	1.07±0.02 ^b	1.33±0.02 ^c	1.07± 0.07 ^b	1.95±0.04 ^f	1.59± 0.02 ^g	1.02± 0.07 ^b
C22:1 (n-9)	n/a	n/a	0.13±0.01	n/a	n/a	n/a	0.10±0.01	n/a
ΣMUFA	27.43± 0.12^a	33.81± 0.11^b	32.44±0.04^d	33.42± 0.04^c	33.43± 0.05^b	33.75± 0.12^g	34.09± 0.13^b	32.99± 0.12^g
C18:2(n-6)	28.91± 0.87 ^a	14.37± 0.05 ^e	18.23± 0.04 ^d	34.44± 0.99 ^c	26.11±0.77 ^b	15.67±0.01 ^e	15.10±0.02 ^a	24.38± 0.04 ^g
C18:2(n-5)	0.66± 0.05 ^a	0.49± 0.02 ^d	0.30± 0.01 ^c	n/a	0.18±0.02 ^b	0.73±0.01 ^f	0.55± 0.01 ^g	n/a
C18:2(n-3)	n/a	n/a	0.81± 0.04 ^b	n/a	0.35± 0.02 ^a	n/a	n/a	n/a
C18:3(n-3)	0.50± 0.02 ^a	0.28± 0.01 ^b	0.45± 0.02 ^a	0.27± 0.04 ^b	0.26± 0.02 ^b	0.35±0.01 ^{bd}	0.34±0.02 ^d	0.28± 0.04 ^b
C20:2(n-6)	0.58± 0.06 ^a	0.38± 0.02 ^c	0.16± 0.02 ^b	0.15±0.01 ^b	0.15±0.01 ^b	n/a	n/a	n/a
C20:3(n-3)	n/a	n/a	n/a	1.01± 0.02 ^a	n/a	n/a	n/a	n/a
C20:4(n-6)	0.26± 0.04 ^a	n/a	1.25± 0.04 ^b	n/a	n/a	0.58±0.02 ^e	0.52±0.03 ^e	n/a
C20:4(n-3)	0.96± 0.04 ^a	0.16± 0.02 ^c	n/a	0.28± 0.01 ^b	n/a	n/a	n/a	0.17± 0.01 ^c
C20:5 (n-3)	1.33± 0.09 ^a	0.76± 0.05 ^e	0.99± 0.05 ^d	0.65± 0.08 ^c	0.40± 0.02 ^b	0.09±0.09 ^f	0.53± 0.01 ^f	1.02±0.07 ^d
C22:6 (n-3)	2.73± 0.15 ^a	4.23±0.09 ^e	7.71±0.09 ^d	2.10± 0.07 ^c	1.84±0.04 ^b	5.89± 0.02 ⁱ	4.83±0.04 ^{hrs}	2.96±0.08 ^f
ΣPUFA	35.94± 0.14^a	20.68± 0.06^c	29.90±0.05^b	31.90± 0.21^a	29.31±0.12^b	23.38± 0.12^g	21.84± 0.02^c	28.81± 0.10^a

Table 1 (Continuation): Effect of smoking processes on the fatty acid composition of *Cyprinus carpio*

	T8'	T9'	T10'	T11'	T12'	T13'	T14'
C12:0	1.20± 0.04 ^b	1.15± 0.04 ^b	1.21±0.02 ^b	0.88± 0.09 ^a	0.89± 0.07 ^a	1.28± 0.04 ^b	0.88± 0.02 ^a
C13:0	0.08± 0.02 ^b	0.07±0.01 ^b	0.05± 0.03 ^a	0.02± 0.01 ^a	0.07±0.01 ^b	n/a	n/a
C14:0	3.29± 0.04 ^c	3.52± 0.08 ^c	3.82± 0.06 ^d	2.26± 0.08 ^a	3.36± 0.08 ^c	2.68±0.03 ^{ba}	2.48± 0.01 ^a
C15:0	0.47±0.02 ^b	0.39±0.07 ^b	0.42± 0.02 ^a	0.40±0.11 ^b	0.39±0.07 ^b	3.24± 0.02 ^f	3.88±0.04 ^f
C16:0	38.28±0.07 ^b	33.80± 0.18 ^a	32.78± 0.14 ^a	36.37±1.01 ^b	31.75±0.18 ^b	33.15± 0.18 ^a	33.21± 0.51 ^a
C17:0	1.94±0.02 ^d	1.13±0.01 ^e	1.23±0.02 ^e	0.53± 0.05 ^b	1.74± 0.04 ^d	n/a	n/a
C18:0	0.68± 0.04 ^a	0.78± 0.05 ^{ac}	0.98±0.04 ^d	0.66± 0.02 ^a	0.84±0.04 ^c	5.89± 0.07 ^f	5.66± 0.44 ^f
C19:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a
C20:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a
C21:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a
ΣSFA	45.94±0.22^d	40.83± 0.06^f	40.44±0.06^f	41.11±0.12^e	39.04± 0.02^g	45.24±0.07^d	46.11± 0.07^d
C16:1(n-7)	n/a	n/a	n/a	n/a	n/a	4.11±0.03 ^e	3.21± 0.02 ^d
C18:1(n-9)	31.44± 0.04 ^b	30.31± 0.41 ^b	31.31± 0.31 ^b	26.53± 0.77 ^a	36.24±0.41 ^d	24.41± 0.18 ^g	25.81± 0.33 ^f
C20:1(n-9)	1.49±0.04 ^e	1.26± 0.04 ^c	1.46± 0.03 ^e	0.46± 0.08 ^d	1.06±0.04 ^b	n/a	n/a
C22:1 (n-9)	0.11± 0.01 ^b	0.26± 0.01 ^a	0.16± 0.02 ^c	n/a	0.24± 0.02 ^a	n/a	n/a
ΣMUFA	33.04± 0.23^g	32.83± 0.04^d	32.93±0.04^f	26.99± 0.10^a	37.54±0.12^e	29.52± 0.04^c	29.02±0.03^c
C18:2(n-6)	14.11±0.05 ^e	18.33± 0.04 ^d	19.01± 0.04 ^d	30.18±1.25 ^f	19.84± 0.03 ^d	16.34±0.22 ^e	15.94±0.19 ^e
C18:2(n-5)	0.65± 0.02 ^a	1.24±0.03 ^e	1.26±0.02 ^e	0.30± 0.05 ^c	1.34± 0.05 ^e	3.41±0.04 ^{hrs}	3.88±0.02 ^{hrs}
C18:2(n-3)	n/a	0.38± 0.05 ^a	0.48±0.03 ^d	n/a	0.51± 0.03 ^c	n/a	n/a
C18:3(n-3)	0.28± 0.01 ^b	0.49± 0.02 ^a	0.39±0.04 ^d	0.09±0.01 ^c	0.31± 0.01 ^b	n/a	n/a
C20:2(n-6)	n/a	n/a	n/a	0.13± 0.02 ^b	n/a	n/a	n/a
C20:3(n-3)	n/a	n/a	n/a	n/a	n/a	0.35± 0.03 ^c	0.24±0.02 ^b
C20:4(n-6)	0.50± 0.04 ^e	1.42±0.04 ^b	1.82±0.05 ^d	n/a	2.25± 0.04 ^c	n/a	n/a
C20:4(n-3)	n/a	n/a	n/a	0.46± 0.07 ^d	n/a	n/a	n/a
C20:5 (n-3)	0.55± 0.03 ^g	0.52± 0.05 ^g	0.28±0.04 ^h	0.10±0.05 ^f	0.72±0.03 ^c	n/a	n/a
C22:6 (n-3)	4.93±0.06 ^{hrs}	3.99± 0.07 ^g	3.49±0.02 ^f	0.61± 0.05 ^f	4.12±0.04 ^e	3.24± 0.03 ^f	4.74±0.04 ^h
ΣPUFA	21.01± 0.02^c	26.37± 0.07^f	26.63± 0.04^f	31.87± 0.02^d	23.41± 0.21^g	24.34±0.08ⁱ	24.80±0.08ⁱ

The value carrying different letters are significantly different ($p < 0.05$) from control and each other when compare all pairs of columns. Results presented are the means of two values followed by their standard deviation. T0' = Control; T1' = Non-eviscerated *Cyprinus carpio* fish smoked with guava tree without filter ; T2' = Non-eviscerated *Cyprinus carpio* fish smoked with guava tree with filter ; T3' = Eviscerated *Cyprinus carpio* fish smoked with guava tree without filter ; T4' = Eviscerated *Cyprinus carpio* fish smoked with guava tree with filter ; T5' = Eviscerated *Cyprinus carpio* fish smoked with mangrove tree without filter ; T6' = Eviscerated *Cyprinus carpio* fish smoked with mangrove tree with filter ; T7' Non-eviscerated *Cyprinus carpio* fish smoked with mangrove tree without filter ; T8' = Non-eviscerated *Cyprinus carpio* fish smoked with mangrove tree with filter ; T9' = Non-eviscerated *Cyprinus carpio* fish smoked with mango tree without filter ; T10' = Non-eviscerated *Cyprinus carpio* fish smoked with mango tree with filter ; T11' = Eviscerated *Cyprinus carpio* fish smoked with mango tree without filter ; T12' = Eviscerated *Cyprinus carpio* fish smoked with mango tree with filter ; T13' = Non-eviscerated *Cyprinus carpio* fish smoked using the traditional method ; T14' = Eviscerated *Cyprinus carpio* fish smoked using the traditional method

T12', T13' and T14' exhibited the highest docosahexaenoic acid contents compared to other samples.

3.2. Effect of smoking on the fatty acid composition of *Claria gariepinus*

The impact of smoking on the fatty acid composition of *Claria gariepinus* is presented in table 2. Results show that the amount of saturated, monounsaturated and polyunsaturated fatty acids was ranged between 38.41-50.39, 27.42 - 33.30 and 19.60-33.39% respectively. Globally, unsaturated fatty acids were higher than saturated fatty acids in this oil. Palmitic acid was the most represented fatty acid (27.27-44.23%) followed by myristic acid (2.13-3.46%). Concerning the monounsaturated fatty acids, oleic acid was the most represented (25.71-31.73%). The major polyunsaturated fatty acid was linoleic acid (13.94-31.70%). This fish was also found to contain more docosahexaenoic acid (0.80-4.46%) than eicosapentaenoic acid (0.00-0.61%). The highest level of docosahexaenoic acid was recorded with T9, T10, T11, T13 and T14. Treatments T1, T2, T5, T6, T8 and T12 significantly ($p < 0.05$) increased the saturated fatty acid content of *Claria gariepinus*. Concerning the content in monounsaturated fatty acids, treatments T2, T4, T6, T8, T9, T10, T13 and T14 significantly increase their concentrations compared to the control and other treatments. Generally, the amount of polyunsaturated fatty acids present in this fish significantly ($p < 0.05$) reduced with the treatments. The highest decrease was recorded with treatments T2, T3, T6, T8 and T12. Treatments T3, T4, T8, T7, T13 and T14 were the best in limiting the loss in polyunsaturated fatty acids.

4. Discussion

Results showed that *Cyprinus carpio* oil contains saturated (34.68 - 46.11%), monounsaturated (26.99 - 34.09%) and polyunsaturated (20.68 - 35.94%) amongst which the total of unsaturated fatty acids was dominant over that of saturated ones. These values were similar to those reported by Ljubojević *et al.* (2016) who showed that *Cyprinus carpio* oil contains 25.38%, 42.99 and 28.37% of saturated fatty acids, monounsaturated fatty acids and polyunsaturated fatty acids respectively. Palmitic (26.54 - 38.28%), oleic (24.41 - 36.24%) and linoleic (14.11 - 34.44%) acids were found to be the most represented fatty acids. The amount of palmitic acid obtained in this study was higher than reported by Ljubojević *et al.* (2016) with the same fish while the proportions of oleic and linoleic acids were found to be within the ranges obtained in this work. *Cyprinus carpio* oil contains higher amount of docosahexaenoic acid (0.61-7.71%) than eicosapentaenoic acid (0.00-1.33%). The presence of these fatty acids in oils even though it was in smaller amount reflects its potential health benefit. In previous studies, docosahexaenoic acid was reported to have good anti-cancer properties (Comba, 2010; Stephenson *et al.*, 2013). In the same line, the antibacterial activity of fish oil rich in omega-3 fatty acids was demonstrated (Hashim *et al.*, 2019; Simplicio *et al.*, 2018). The presence of linoleic acid which is an essential fatty acid also demonstrates the potential good biological property of *Cyprinus carpio* oil (Dobryniewski *et al.*, 2007). Results also showed that all treatments significantly increase the saturated and monounsaturated fatty acid content of *Cyprinus carpio* oil compared to the control while they reduced the amount of polyunsaturated fatty acids. An increase in saturated and monounsaturated during smoking was reported

by Ljubojević *et al.* (2016) with the same fish and by Salaudeen *et al.* (2018) with *Clarias gariepinus* Burchell, 1822. However, the decrease in polyunsaturated fatty acids registered after smoking is not in line with the findings of these authors. The decrease can be attributed to the oxidation of polyunsaturated fatty acids during treatment, since high temperature was used throughout the process. It has been reported that, when food samples are subjected to high temperature, oxidation reactions can be initiated and lead to the formation of hydroperoxides which are primary oxidation products, and to aldehydes and ketones which are secondary oxidation products responsible of off-flavours in oils and fats (Bansal *et al.*, 2010). The formation of these products subsequently causes the destruction of the normal fatty acid. Treatments T1', T2', T5', T6', T8', T9', T10', T12', T13' and T14' presented the highest docosahexaenoic acid contents compared to the control and samples obtained from other treatments. This suggests that they are suitable for the preservation of docosahexaenoic acid of which the beneficial biological properties were demonstrated. These results are in agreement with those of Ljubojević *et al.* (2016) who obtained docosahexaenoic acid values of 1.19 and 1.42% respectively in *Cyprinus carpio* oil. T3' and T11' exhibited significantly higher linoleic acid content compared to the control and other treated samples. This result corroborate with the findings of Olopade & Dienye (2023) who reported that hot and cold smoking significantly increase the linoleic acid content of fish.

Concerning the effect of smoking on the fatty acid composition of *Claria gariepinus* oil, results revealed saturated, monounsaturated and polyunsaturated fatty acids contents ranged between 38.41-50.39, 27.42 - 33.30 and 19.60-

33.39% respectively. The range of saturated fatty acids was higher than that 25.91-32.29% reported by Ogunbambo (2020). However, the monounsaturated and polyunsaturated fatty acids concentrations were lower than 44.12-45.93 and 29.6-33.76% respectively obtained by this author. The smoking methods as well as the extraction procedure can explain the differences observed. As previously noticed with the *Cyprinus carpio* oil, the total unsaturated fatty acids were higher than the total saturated ones. Similar results were obtained by Salaudeen *et al.* (2018) and Olopade & Dienye (2023) with the oil from the raw and smoked *Claria gariepinus*. Palmitic (27.27-44.23%), oleic (25.71-31.73%) and linoleic (13.94-31.70%) acids were found to be the most represented fatty acids. However, the concentrations in these fatty acids were higher than the values obtained by Olopade & Dienye (2023) with the oil from the same fish. The smoking conditions, oil extraction methods as well as the environment in which the fish was captured can explain the differences observed (Rora *et al.*, 1999). Docosahexaenoic (0.80-4.46%) was present in higher amount compared to eicosapentaenoic acid (0.00-0.61%). The presence of these two fatty acids in *Claria gariepinus* oil obtained in this study is in line with the findings of Salaudeen *et al.* (2018) who obtained docosahexaenoic acid and eicosapentaenoic acid contents of 2.31-4.25 and 0.86-3.33% respectively with the same fish in Nigeria. These fatty acids were present in higher amounts in oils from Treatments T9, T10, T11, T13 and T14 gave the highest amounts of docosahexaenoic acid while T1, T2, T5, T6, T8 and T12 increased the saturated fatty acid content of *Claria gariepinus* oil. Treatments T9, T10, T11, T13 and T14 should therefore be encouraged to be applied for a good extraction of docosahexaenoic acid and eicosapentaenoic acid.

Table 2: Effect of smoking on the fatty acid composition of *Clarias gariepinus*

	T0	T1	T2	T3	T4	T5	T6	T7
C12:0	0.76± 0.06 ^a	0.88± 0.08 ^a	0.89± 0.01 ^a	1.03± 0.04 ^b	1.01± 0.05 ^b	1.31± 0.11 ^c	1.26± 0.09 ^c	0.91± 0.04 ^a
C13:0	0.02± 0.00 ^a	0.04± 0.02 ^a	0.04± 0.03 ^a	0.03± 0.01 ^a	0.03± 0.01 ^a	0.03± 0.00 ^a	0.03± 0.02 ^a	0.03± 0.00 ^a
C14:0	2.64± 0.12 ^a	2.61± 0.08 ^b	2.42± 0.05 ^c	2.25± 0.10 ^c	2.81±0.99 ^b	2.66±0.06 ^b	2.58± 0.54 ^b	2.44± 0.05 ^c
C15:0	0.36± 0.04 ^a	0.35± 0.02 ^a	0.55± 0.02 ^b	0.41±0.01 ^c	0.52±0.06 ^b	0.41± 0.07 ^c	0.22± 0.07 ^c	0.26± 0.01 ^c
C16:0	43.13± 2.01 ^a	43.78± 0.49 ^a	40.38± 0.07 ^c	33.66±0.42 ^b	43.39± 2.10 ^a	38.58± 0.77 ^e	40.51± 2.01 ^c	35.24±0.74 ^d
C17:0	0.56± 0.04 ^a	0.48± 0.01 ^c	0.47± 0.04 ^c	0.56± 0.05 ^a	0.69± 0.06 ^b	0.35± 0.08 ^d	0.83±0.04 ^d	0.59± 0.03 ^a
C18:0	0.94± 0.10 ^a	0.67± 0.07 ^c	0.45± 0.01 ^b	0.43±0.06 ^b	0.89± 0.05 ^a	0.58±0.07 ^e	0.63±0.08 ^c	0.29± 0.04 ^d
C19:0	n/a	n/a	0.13± 0.02 ^a	n/a	n/a	n/a	n/a	n/a
C20:0	n/a	n/a	0.03± 0.02 ^a	0.04± 0.02 ^a	0.05± 0.01 ^a	0.07±0.01 ^b	n/a	n/a
C21:0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
ΣSFA	38.41± 0.31^a	48.81±0.11^b	45.36± 0.08^d	38.41± 0.08^c	39.39± 0.41^b	43.99±0.14^e	46.06± 0.29^d	39.76± 0.11^c
C16:1(n-7)	n/a	0.38± 0.02 ^b	0.09± 0.01 ^a	n/a	0.11± 0.02 ^a	0.12± 0.04 ^a	0.41±0.07 ^b	0.19± 0.02 ^c
C18:1(n-9)	27.67± 1.20 ^a	27.07± 0.66 ^a	31.61±1.02 ^b	27.99± 0.22 ^a	31.82± 1.02 ^b	26.86± 0.40 ^c	32.23±0.88 ^b	27.21± 1.02 ^a
C20:1(n-9)	0.53± 0.04 ^a	0.45± 0.02 ^b	0.46±0.04 ^b	0.46±0.04 ^b	0.45± 0.02 ^b	0.44±0.08 ^b	0.45±0.07 ^b	0.63± 0.07 ^c
C22:1 (n-9)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	0.02±0.01
ΣMUFA	28.20± 0.80^a	27.91± 0.12^a	32.16±0.21^b	28.45± 0.11^a	32.38±0.02^b	27.42± 0.14^a	33.09± 0.24^c	28.02± 0.24^a
C18:2(n-6)	21.18± 0.99 ^a	21.44± 0.08 ^a	19.03±0.05 ^d	31.70± 1.02 ^c	15.92±0.75 ^b	26.58± 0.42 ^e	18.46± 0.44 ^d	30.17± 0.58 ^c
C18:2(n-5)	0.47± 0.04 ^a	0.09±0.01 ^b	0.42± 0.02 ^a	n/a	0.46± 0.10 ^a	0.41± 0.02 ^b	n/a	0.38± 0.01 ^c
C18:2(n-3)	0.07± 0.02 ^a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
C18:3(n-3)	n/a	n/a	n/a	0.12± 0.03 ^a	0.14± 0.02 ^a	0.15± 0.01 ^a	0.96± 0.07 ^c	0.07±0.01 ^b
C20:2(n-6)	0.11± 0.04 ^a	0.28± 0.02 ^c	n/a	0.02±0.01 ^b	0.03± 0.00 ^b	n/a	0.02±0.01 ^b	n/a
C20:3(n-3)	n/a	n/a	n/a	0.03± 0.01 ^b	0.10± 0.01 ^a	0.06±0.02 ^b	n/a	n/a
C20:4(n-6)	0.74± 0.04 ^a	n/a	1.72±0.12 ^b	n/a	n/a	n/a	n/a	n/a
C20:4(n-3)	0.04± 0.00 ^a	n/a	n/a	n/a	0.07± 0.03 ^a	0.04± 0.01 ^a	0.08± 0.01 ^a	0.73± 0.03 ^b
C20:5 (n-3)	0.25± 0.07 ^a	0.18± 0.01 ^b	0.30± 0.01 ^a	0.29± 0.02 ^a	0.20± 0.04 ^b	0.17± 0.02 ^b	0.20± 0.05 ^b	0.05±0.01 ^c
C22:6 (n-3)	1.27± 0.12 ^a	1.26± 0.07 ^a	1.00±0.07 ^b	0.94±0.01 ^b	1.32± 0.14 ^a	1.20± 0.13 ^a	1.11± 0.10 ^a	0.80±0.04 ^c
ΣPUFA	33.39± 0.10^a	23.28± 0.02^a	22.48± 0.10^a	33.13± 0.11^c	28.27± 0.08^b	28.59± 0.07^d	20.85±0.10^f	32.22± 0.11^c

Table 2 (Continuation): Effect of smoking on the fatty acid composition of *Clarias gariepinus*

	T8	T9	T10	T11	T12	T13	T14
C12:0	0.88± 0.05 ^a	1.33± 0.04 ^c	1.59± 0.05 ^d	1.03± 0.01 ^b	0.98± 0.10 ^b	0.98± 0.04 ^b	1.08± 0.02 ^b
C13:0	0.03± 0.01 ^a	0.05± 0.01 ^a	0.07± 0.01 ^b	n/a	n/a	n/a	n/a
C14:0	2.55± 0.07 ^b	3.46± 0.0 ^d	3.36± 0.08 ^d	2.13± 0.04 ^b	3.19± 0.33 ^d	2.23± 0.02 ^b	2.38± 0.04 ^b
C15:0	0.30± 0.02 ^a	0.39± 0.07 ^a	0.39± 0.07 ^a	0.51± 0.01 ^b	0.53± 0.09 ^b	2.22± 0.02 ^f	3.71± 0.06 ^e
C16:0	40.12± 0.10 ^c	31.72± 0.18 ^b	33.92± 0.18 ^b	35.07± 0.06 ^d	44.23± 1.01 ^a	31.25± 0.11	27.27± 0.08
C17:0	0.43± 0.02 ^c	1.66± 0.05 ^e	1.56± 0.04 ^e	0.50± 0.02 ^c	0.84± 0.05 ^d	n/a	n/a
C18:0	0.57± 0.07 ^e	0.74± 0.03 ^c	0.72± 0.03 ^c	0.77± 0.02 ^f	0.62± 0.01 ^c	5.11± 0.08 ^b	6.84± 0.04 ^b
C19:0	0.09± 0.01 ^b	n/a	n/a	n/a	n/a	n/a	n/a
C20:0	0.04± 0.02 ^a	n/a	n/a	0.19± 0.05 ^c	n/a	n/a	n/a
C21:0	n/a	n/a	n/a	0.81± 0.02 ^a	n/a	n/a	n/a
ΣSFA	45.01± 0.04^d	39.35± 0.07^c	41.61± 0.09^e	41.01± 0.12^e	50.39± 0.26^f	41.79± 0.05^e	41.28± 0.07^e
C16:1(n-7)	0.33± 0.01 ^d	n/a	n/a	4.61± 0.01 ^e	n/a	3.99± 0.02 ^e	4.91± 0.04 ^e
C18:1(n-9)	32.45± 1.02 ^b	31.73± 0.41 ^b	30.73± 0.41 ^b	25.71± 0.86 ^d	29.28± 0.07 ^a	29.21± 0.08 ^a	28.21± 0.12 ^a
C20:1(n-9)	0.52± 0.03 ^a	1.06± 0.04 ^e	1.06± 0.04 ^e	0.08± 0.04 ^f	0.73± 0.05 ^d	n/a	n/a
C22:1 (n-9)	n/a	0.28± 0.03 ^a	0.26± 0.01 ^a	0.70± 0.01 ^b	n/a	n/a	n/a
ΣMUFA	33.30± 0.18^c	33.07± 0.14^c	32.05± 0.12^b	31.10± 0.21^e	30.01± 0.06^d	33.20± 0.14^c	33.12± 0.06^c
C18:2(n-6)	19.13± 0.05 ^d	17.97± 0.07 ^f	17.44± 0.05 ^f	13.94± 0.04 ^e	15.29± 0.44 ^b	15.34± 0.11 ^b	15.34± 0.11 ^b
C18:2(n-5)	n/a	1.41± 0.04 ^d	1.54± 0.02 ^d	2.99± 0.10 ^e	0.37± 0.04 ^c	3.64± 0.08 ^f	3.64± 0.08 ^f
C18:2(n-3)	n/a	0.44± 0.04 ^b	0.47± 0.02 ^b	0.51± 0.05 ^b	n/a	n/a	n/a
C18:3(n-3)	0.98± 0.02 ^c	0.34± 0.03 ^d	0.29± 0.01 ^d	1.02± 0.03 ^e	0.21± 0.01 ^d	n/a	n/a
C20:2(n-6)	0.06± 0.01 ^a	n/a	n/a	1.18± 0.06 ^c	0.09± 0.01 ^a	n/a	n/a
C20:3(n-3)	n/a	n/a	n/a	2.80± 0.07 ^c	n/a	2.94± 0.06 ^c	2.94± 0.06 ^c
C20:4(n-6)	n/a	2.32± 0.04 ^e	2.06± 0.04 ^c	0.33± 0.01 ^d	n/a	n/a	n/a
C20:4(n-3)	0.04± 0.02 ^a	n/a	n/a	0.95± 0.04 ^d	1.79± 0.07 ^c	n/a	n/a
C20:5 (n-3)	0.31± 0.01 ^a	0.59± 0.04 ^d	0.61± 0.02 ^d	0.06± 0.01 ^c	0.35± 0.04 ^a	n/a	n/a
C22:6 (n-3)	1.15± 0.05 ^a	4.02± 0.05 ^e	3.97± 0.05 ^e	4.46± 0.02 ^e	1.51± 0.08 ^d	3.64± 0.02 ^e	3.64± 0.02 ^e
ΣPUFA	21.69± 0.08^e	27.59± 0.11^d	26.34± 0.08^e	27.89± 0.12	19.60± 0.05^b	25.56± 0.08^e	25.60± 0.08^e

The value carrying different letters are significantly different ($p < 0.05$) from control and each other when compare all pairs of columns. Results presented are the means of two values followed by their standard deviation. T0 = Control ; T1 = Non-eviscerated *Clarias gariepinus* fish smoked with guava tree without filter; T2 = Non-eviscerated *Clarias gariepinus* fish smoked with guava tree with filter; T3 = Eviscerated *Clarias gariepinus* fish smoked with guava tree without filter; T4 = Eviscerated *Clarias gariepinus* fish smoked with guava tree with filter ; T5 = Non-eviscerated *Clarias gariepinus* fish smoked with Mangrove tree without filter ; T6 = Non-eviscerated *Clarias gariepinus* fish smoked with Mangrove tree with filter ; T7 = Eviscerated *Clarias gariepinus* fish smoked with Mangrove tree without filter ; T8 = Eviscerated *Clarias gariepinus* fish smoked with Mangrove tree with filter ; T9 = Non-eviscerated *Clarias gariepinus* fish smoked with Mango tree without filter ; T10 = Non-eviscerated *Clarias gariepinus* fish smoked with mango tree with filter ; T11 = Eviscerated *Clarias gariepinus* fish smoked with mango tree with filter ; T12 = Eviscerated *Clarias gariepinus* fish smoked with Mango tree with filter ; T13 = Non-eviscerated *Clarias gariepinus* fish smoked using the traditional method ; T14 = Eviscerated *Clarias gariepinus* fish smoked using the traditional method

The health benefit of these fatty acids against non-communicable diseases was reported (Hashim *et al.*, 2019; Simplicie *et al.*, 2018). However, treatments T2, T4, T6, T8, T9, T10, T13 and T14 significantly increased the monounsaturated fatty acid content of this oil. The concentrations of polyunsaturated fatty acids were found to significantly decrease with the treatments and were severe with treatments T2, T3, T6, T8 and T12. The treatment conditions might explain the drastic drop in polyunsaturated fatty acids. Treatments T2, T3, T6, T8 and T12 should therefore be avoided for a better preservation of polyunsaturated fatty acids. As previously mentioned with *Cyprinus carpio* oil, the decrease in polyunsaturated fatty acids observed can be attributed to the oxidation of polyunsaturated fatty acids during the treatment, since high temperature was used throughout the process (Bansal *et al.*, 2010). T2 and T7 presented significantly higher linoleic acid content compared to the control and other treated samples. This result corroborate with the findings of Salaudeen *et al.* (2018) who reported that smoking significantly increase the linoleic acid content of oil from *Clarias gariepinus* Burchell, 1822.

5. Conclusion

The objective of this study was to evaluate the effect of different smoking methods on the fatty acid composition of oils extracted from *Clarias gariepinus* and *Cyprinus carpio*. Results showed that all the smoking methods significantly increase the saturated and monounsaturated fatty acid content of these fish while they decrease the polyunsaturated fatty acid content. Treatments T2, T7, T3' and T11' are the best for a good content in linoleic acid. Treatments T1' T2', T5', T6' T8', T9', T10', T12', T13' and T14' are the best for a good content in docosahexaenoic acid in the oil from *Cyprinus carpio* while treatments

T9, T10, T11, T13 and T14 were the best in docosahexaenoic acid extraction from *Clarias gariepinus*. These treatments should be encouraged for smoking of these fish in order to gain more in polyunsaturated fatty acids which have been proven to be efficient in the prevention and management of non-communicable diseases.

Conflict of interest

The authors declare that they have no competing interests.

Ethics

This Study does not involve Human or Animal Testing.

Acknowledgement

None

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Cite this paper as: Tiwo, T.C., Djimbie, D.J., Ndomou, H.S.C.R., Sidi, K.D., Nayak, B.B., Womeni, H.M., & Tchoumboungang, F. (2024). Effect of Different Smoking Methods on the Fatty Acid Composition of Oils Extracted from Two Fish (*Cyprinus carpio* and *Clarias gariepinus*) Captured from Fish Ponds, Batié, Cameroon. *Journal of Food Stability*, 7(1), 49-61.
[DOI: 10.36400/J.Food.Stab.7.1.2024-004](https://doi.org/10.36400/J.Food.Stab.7.1.2024-004)